Sampling Streets: Design-Based Inference for Street-Level Field Measurement

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Abstract

Researchers measure streets with cameras and sensors: the condition of roads, who is visible in public, how loud and how polluted the street is. The data usually come from wherever access was easy, so the numbers describe the places measured and not the city. We describe what it takes to make such a measurement a statement about a city: a list of road segments to draw from, a random draw, and standard errors that follow from the draw. We then show that drawing points at random is not enough. When the observed unit is the drawn unit, as when a Street View image is requested at a sampled segment, the design carries over and only coverage needs care. When a field team rides between sampled points with a camera or a meter, the observations describe the route, not the sample: routes run on bigger roads than the points sit on, and a camera cut at fixed time intervals photographs slow stretches more often. In Delhi, 85% of sampled points but 25% of video frames are on residential roads. Weighting frames by speed and reweighting road classes to the drawn sample moves the estimated share of women among pedestrians in Delhi from 19.8% to 23.5%, and the share of noise readings peaking at or above 100 dB from 14.4% to 8.1%. In a simulation with a known truth, intervals from the unadjusted route sample cover the truth 32% of the time and the adjusted ones 95%. We apply the framework to road condition in four cities, women's presence on the street in four Indian cities, and noise and air quality in Delhi.

*Data and code to reproduce every number in this paper are at https://github.com/geosensing/streetscan. The sampler, the route allocator, and the estimation package are at https://github.com/geosensing.

1 Introduction

Measurements taken on the street are an increasingly common source of data in the social sciences. Researchers code neighborhood disorder, road and sidewalk condition, and the composition of public space from street-level imagery [Rundle et al., 2011, Kelly et al., 2013, Naik et al., 2014, Gebru et al., 2017], and they measure noise, particulate matter, and temperature with sensors. Most of this work draws on data that are convenient: a platform's image corpus, a census of named blocks, monitors at fixed stations, or routes a team could reach. Convenient data deliver volume. They do not say what population the numbers describe, and standard errors computed from them treat observations as independent draws from an unspecified population.

Survey sampling has a standard answer. List the units of the population, draw some at random with known probabilities, and weight each observation by the inverse of its probability of being drawn [Horvitz and Thompson, 1952, Cochran, 1977, Lohr, 2019]. For streets, the list can be built from the road network: OpenStreetMap roads [OpenStreetMap contributors, 2026] cut into short segments. We built tools to do this: a sampler that cuts roads into segments and draws from them, an allocator that groups sampled points into routes a field team can ride in a day, and an estimation package.

This paper makes two points. The first is that the standard machinery applies directly, and easily, when the unit observed is the unit drawn. If an image is requested at the start of each sampled segment, each image stands for the segments the draw says it stands for. The remaining problem is coverage: some sampled segments have no image.

The second point is a correction to our own practice. When a field team rides between sampled points with a camera that is always recording, or stops to read a meter where it is convenient to stop, the observations are not at the sampled points. The sample fixes a route, and the data describe the route. A route differs from the road network in two ways that the data can show and partly repair. Routes between points run on bigger roads than the points sit on. And a camera that saves a frame every few seconds photographs a stretch of road in proportion to the time spent on it, so slow stretches, which are also the crowded ones, are over-represented. We work out what each of these does, give estimators that undo them, check the estimators in a simulation with a known truth, and report how much the corrections move estimates in practice.

We use five applications. Road condition in Bangkok, Jakarta, Lagos, and Wayne County, Michigan, comes from Street View images at sampled segments. Women's share of pedestrians comes from Street View at sampled locations in Mumbai, Navi Mumbai, and Delhi, and from video recorded on a motorbike in those cities and Bangalore, an outcome motivated

by the concern that women are underrepresented in everyday public life [Sen, 1990, Phadke et al., 2011]. Noise and air quality in Delhi come from hand-held meters read along the same routes; road-traffic noise affects sleep, cardiovascular health, and cognition [World Health Organization. Regional Office for Europe, 2018, World Health Organization, 2022].

2 Estimands

Let $i \in \mathcal{U} = \{1, \ldots, N\}$ index the units of a finite population, here segments of the road network. For unit i, Y_i is a measurement and, where relevant, Z_i a denominator. For counts of people by gender, Y_i is the number of women visible and Z_i the number of people classified as women or men. For a sound-meter reading, Y_i is a level in decibels and there is no denominator.

The *population ratio* is

$$R = \frac{\sum_{i \in \mathcal{U}} Y_i}{\sum_{i \in \mathcal{U}} Z_i}. \tag{1}$$

For images, R is the share of visible people who are women. A market with two hundred people counts a hundred times as much as a side street with two.

The *population mean* is $\mu = N^{-1} \sum_{i \in \mathcal{U}} Y_i$: for noise, the sound level at a segment drawn at random. The *threshold share* is

$$P_\tau = \frac{1}{N} \sum_{i \in \mathcal{U}} \mathbb{1}\{Y_i \geq \tau\}, \tag{2}$$

the share of segments at or above a level τ. For noise we use $\tau = 55$ dB, the daytime ambient standard for residential areas under India's Noise Pollution Rules [Ministry of Environment and Forests, Government of India, 2000], for the quietest moment in a reading, and $\tau = 100$ dB for the loudest.

A fourth quantity, the *location mean* of a per-unit ratio,

$$L = \frac{1}{N^+} \sum_{i \in \mathcal{U}: Z_i > 0} \frac{Y_i}{Z_i}, \tag{3}$$

counts a side street the same as a market. R below L would mean that crowded places are more male.

All four are defined over segments. Whether data estimate them depends on how the data were collected, which is the subject of the next section. A camera that saves a frame every minute estimates what a rider sees per minute. That is a well-defined quantity, but it is not R.

3 Two Designs

3.1 The frame

The frame is a list of road segments. Roads of chosen classes are taken from OpenStreetMap and cut into pieces of about 500 meters; each piece is a row. The frame says what the estimate is about. A frame of primary, secondary, tertiary, and trunk roads supports claims about major roads. A frame that adds residential streets supports claims about the network people live on, most of which is residential: in the frames used here, residential streets are 60 to 85% of segments.

The simplest draw takes n segments from N with equal probability, so each has inclusion probability $\pi_i = n/N$. A stratified draw takes n_h of the N_h segments in each road class h, so $\pi_i = n_h/N_h$.

3.2 Point samples

In a point sample the unit observed is the unit drawn. The clearest case is platform imagery. Google Street View's metadata service says whether a panorama exists near a coordinate and when it was taken, and its image service returns the picture [Google, 2026b,c]. Query it at the start of each sampled segment, and each image inherits its segment's π_i.

What can go wrong is coverage. Some sampled segments have no panorama, and not every available image gets coded. If the missing segments differ from the rest, estimates describe covered streets only. The share of sampled segments with an image is itself an estimate of the platform's coverage of the frame, and it should be reported.

3.3 Route samples

Sending a person to each sampled point is expensive, so sampled points are grouped into itineraries and a rider travels each itinerary. In our fieldwork the rider wore a camera that recorded continuously, and frames were later saved from the video at fixed intervals: every 10 seconds in Mumbai and Navi Mumbai and every 120 seconds in Bangalore and Delhi. No frame is tied to a sampled point. The median frame is 68 meters from the nearest sampled point in Mumbai, 368 in Delhi, and 450 in Bangalore. Sound-meter readings were taken at stops along the route, a median 112 meters from the nearest sampled point, and 45% of them within 100 meters of one.

The sampled points therefore fix the route, and the observations are a sample of the route. Two things separate them from the frame.

Time. A frame every t seconds lands on a given stretch of road with probability proportional to the time the rider spends there, which is the stretch's length divided by the rider's speed. A stretch ridden at 5 km/h is photographed five times as often as one ridden at 25 km/h. Speed is lowest where streets are narrow or crowded, so the frames over-represent those places relative to distance. In Delhi the median speed at a frame is 8 km/h on residential roads and 26 km/h on secondary and larger roads, and in 14% of frames the rider is stationary. Weighting each frame by the rider's speed at that moment, v_i, turns a sample that is uniform in time into one that is uniform in distance:

$$\widehat{R}_{\text{route}} = \frac{\sum_{i \in \mathcal{S}} v_i Y_i}{\sum_{i \in \mathcal{S}} v_i Z_i}. \tag{4}$$

Speed comes from the camera's GPS track. A stationary frame would get no weight, so we floor speed at 0.5 meters per second.

Roads. The quickest way between two points on residential streets usually runs along a larger road. Figure 1 compares the road classes of the sampled points with those of the frames. In Delhi, 85% of sampled points and 25% of frames are on residential roads; 4% of points and 42% of frames are on secondary or larger roads. Weighting by speed makes this worse, not better, because it shifts weight toward fast roads. The repair is post-stratification: rescale the weights so that each road class has the share it has in the frame. The drawn sample gives those shares. This removes the imbalance across road classes. It cannot remove differences within a class between the streets a route uses and the streets it does not, a limit we return to.

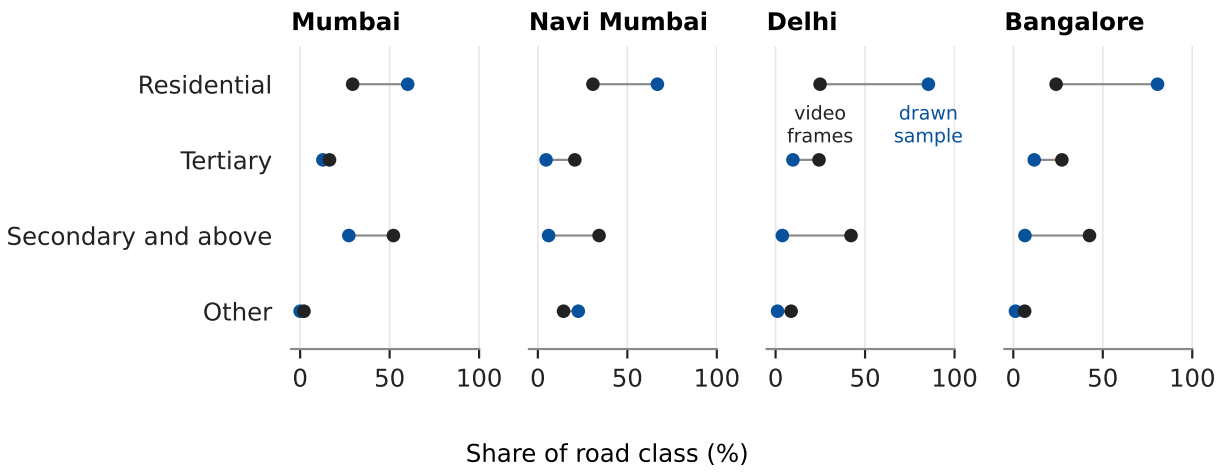


Figure 1: Road classes of the sampled points and of the video frames recorded while riding between them. Each panel is a city. Blue points give the share of the 1,000 sampled points in each OpenStreetMap road class; black points give the share of annotated video frames, classed by the nearest mapped road. "Other" includes service roads, footways, unclassified roads, and frames with no mapped road nearby.

Which of the three quantities is wanted depends on the question. Per minute of riding is the rider's experience. Per meter of route describes the roads ridden. Reweighted to the frame is the closest a route sample comes to the road network.

3.4 Allocation

Grouping sampled points into itineraries does not change which points were drawn. It does make observations depend on one another: frames from one day share weather, a time of day, a part of the city, and a rider. Standard errors must treat the day or the itinerary, not the frame, as the independent unit.

Our allocator can also generate random walks on the road graph in place of routes between sampled points. A simple random walk on an undirected graph crosses every edge equally often in the long run, so measurements taken at a fixed distance along it would weight roads by length without any reweighting. That property is a limit: it holds for long walks and fails on one-way streets. None of the applications here used walks, and we do not evaluate them.

4 Estimation

4.1 Weighted totals, ratios, and means

Let w_i be the weight of sampled unit i. For a point sample, $w_i = 1/\pi_i$. The Horvitz–Thompson estimator of a total,

$$\widehat{T}_Y = \sum_{i \in \mathcal{S}} w_i Y_i, \tag{5}$$

is unbiased for $T_Y = \sum_{i \in \mathcal{U}} Y_i$ over repeated draws [Horvitz and Thompson, 1952]. The ratio and the mean are estimated by

$$\widehat{R} = \frac{\sum_{i \in \mathcal{S}} w_i Y_i}{\sum_{i \in \mathcal{S}} w_i Z_i}, \qquad \bar{Y} = \frac{\sum_{i \in \mathcal{S}} w_i Y_i}{\sum_{i \in \mathcal{S}} w_i}. \tag{6}$$

The threshold share is the mean of $\mathbb{1}\{Y_i \geq \tau\}$. With equal-probability sampling the weights cancel and these are the sample ratio and the sample mean.

Proposition 1 (Ratio estimator)**.** *If every unit has a known positive inclusion probability and measurement returns fixed values (Y_i, Z_i), then $\widehat{R}$ is consistent for R with bias of order n^{-1}, and its variance is estimated to first order by the variance of the weighted total of $e_i = (Y_i - \widehat{R} Z_i)/\widehat{T}_Z$.*

The bias arises because $\widehat{R}$ divides one random total by another [Cochran, 1977].

4.2 Variance

Write $c_i = w_i e_i$ for a ratio and $c_i = w_i (Y_i - \bar{Y}) / \sum_j w_j$ for a mean. Sum c_i within each independent unit g (a segment in a point sample, a day or an itinerary in a route sample) to get C_g. With G_h independent units in stratum h,

$$\widehat{\mathbb{V}} = \sum_h \frac{G_h}{G_h - 1} \sum_{g \in h} \left(C_g - \bar{C}_h \right)^2. \tag{7}$$

With one stratum and each row its own unit, this is the usual variance of a ratio or a mean under simple random sampling; with days or itineraries as units, it is the cluster-robust variance [Cameron and Miller, 2015]. It treats units as drawn with replacement and so omits the finite population correction. That overstates the variance by the sampling fraction. The fraction is about 1% or less in the Street View samples of Indian cities. It may be as large as a sixth in the major-road samples (Bangkok's 1,000 sampled segments carry identifiers up to 6,267), so those standard errors are somewhat conservative. We omit the correction

throughout because the size of the frame was not saved with most of the samples; the sampler now saves it.

After post-stratification, c_i is replaced by its departure from the weighted mean of c in its road class. Fixing the class shares removes the variation between classes, and only variation within classes remains.

Intervals use a t distribution with degrees of freedom equal to the number of independent units minus the number of strata. The video samples have 9 to 14 days each, and intervals based on so few clusters tend to be too narrow [Cameron and Miller, 2015].

All of this is implemented in the `geoestimate` package. Its weighted, stratified, clustered, and post-stratified estimates and standard errors agree with those of R's `survey` package to eight digits on test data.

4.3 Coverage

A sample is complete only when every drawn unit is either observed or recorded as missing. If units are missing at different rates in different road classes, post-stratifying the observed units to the drawn sample's class shares corrects for it, on the assumption that within a class the missing units resemble the observed ones. If that assumption fails, the estimate describes covered streets.

5 A Check Against a Known Truth

We built an artificial city of 20,000 segments in three road classes. Classes differ in women's share of pedestrians, in crowding, and in speed, and within a class more crowded segments are slower. The true share of women among pedestrians is 24.7%. We drew two kinds of sample a thousand times each. The point sample observes 600 segments drawn at random. The route sample rides 150 segments on each of 12 days, half of them on major roads though major roads are 8% of the city, and saves a frame every 60 seconds; each day also shifts the share of women a little, so frames from one day are correlated.

Table 1: Estimates of women's share of pedestrians in a simulated city where the true share is 24.7%. Each row summarizes 1,000 samples. "Per minute" weights frames equally; "per metre" weights each frame by speed; "road network" also post-stratifies road classes to their shares in the city. Mean and SD are the mean and standard deviation of the estimates across samples, bias is the mean minus the truth, SE is the mean estimated standard error, and coverage is the share of nominal 95% intervals that contain the truth.

	Mean (%)	Bias (pp)	SD (pp)	SE (pp)	Coverage (%)
Point sample	24.7	0.0	1.4	1.4	95
Route sample, per minute	22.2	−2.6	1.0	0.9	32
Route sample, per metre	19.9	−4.9	1.0	0.9	1
Route sample, road network	24.9	+0.2	1.1	1.1	95

Table 1 gives the results. The point sample is unbiased and its intervals cover the truth 95% of the time. The route sample taken at face value is 2.6 points too low and its intervals cover 32% of the time. Weighting by speed alone is worse, 4.9 points too low, because it moves weight from slow residential streets to fast major roads, which the route already over-represents. Speed weights and post-stratification together remove the bias, and the intervals cover 95% of the time.

The simulation gives post-stratification its best case: road class is the only way the route differs from the city. Real routes may also favor particular streets within a class.

6 Applications

6.1 Road condition from Street View

This is the point sample in its plainest form. In late 2017 we drew 1,000 segments each of primary, secondary, tertiary, and trunk road in Bangkok, Dhaka, Jakarta, and Lagos, and 5,000 in Wayne County, Michigan, and requested Street View images at the start of each. Workers on Amazon Mechanical Turk coded each image for potholes, cracks, litter, a paved sidewalk, and clear road markings.

Table 2: Condition of major roads from Street View images at sampled road segments. Entries are percentages of segments. The frame is primary, secondary, tertiary, and trunk roads cut into 500-meter segments; "Covered" is the share of sampled segments for which Street View returned an image. Estimates are post-stratified to the drawn sample's mix of road classes.

	Sampled	Covered (%)	Potholes	Cracks	Litter	Paved sidewalk	Clear markings
Bangkok	1,000	100	6	24	6	51	83
Jakarta	1,000	98	23	44	21	49	35
Lagos	1,000	87	6	20	15	30	24
Wayne County	5,000	97	7	62	9	67	66

Coverage was 24.6% in Dhaka, which we therefore dropped, 87% in Lagos, and at least 97% elsewhere (Table 2). 23% of segments in Jakarta show potholes, against 6% to 7% in the other three places; standard errors across the table run from 0.4 to 1.6 percentage points. Adjusting for the road classes of the uncovered segments changes no estimate by more than 0.6 points. The estimates are about the streets as they were when photographed, which ranged over several years.

6.2 Women's presence from Street View

The second point sample uses a frame that adds residential and unclassified roads. In each of Mumbai, Navi Mumbai, and Delhi, 1,500 sampled locations were checked for Street View coverage; 83%, 76%, and 78% had a panorama. Coders counted women and men on foot in images from 498, 437, and 500 locations. How those locations were chosen from the covered ones was not recorded, so the estimates rest on the assumption that the choice was unrelated to who was on the street.

Women are 20.8% of pedestrians in Mumbai, 25.0% in Navi Mumbai, and 20.2% in Delhi (Table 3, last column). Adjusting for road class moves each by less than half a point.

Platform imagery has properties that matter for an outcome like this one. Google blurs faces and license plates [Google, 2026a, Frome et al., 2009], which protects the people photographed and makes some of them harder to classify. The platform, not the researcher, chooses when to photograph, and its vehicles work in daylight. And images can be years old, with dates that vary within a city [Curtis et al., 2013]. Street View is better suited to things that change slowly, such as sidewalks, than to who is on the street at a given hour.

6.3 Women's presence from video

A rider with a chest-mounted camera rode itineraries through 1,000 sampled points in each of Mumbai, Navi Mumbai, Bangalore, and Delhi over 11, 9, 13, and 14 days. Coders counted

women and men on foot in 9,240 frames. This is a route sample, and Table 3 and Figure 2 report it three ways.

Table 3: Women's share of pedestrians (%), with 95% confidence intervals. The first three columns use video frames. "Per minute of riding" weights frames equally; "per metre of route" weights each frame by the rider's speed; "road network" also post-stratifies road classes to the drawn sample. Their intervals treat fieldwork days as clusters. "Street View" uses images at sampled locations, post-stratified to the drawn sample's road classes, with locations as independent units. Street View was not collected in Bangalore.

	Per minute of riding	Per metre of route	Road network	Street View
Mumbai	19.1 [17.4, 20.8]	19.0 [17.4, 20.6]	20.1 [17.9, 22.2]	20.8 [18.8, 22.9]
Navi Mumbai	18.0 [15.7, 20.4]	17.8 [15.8, 19.7]	18.9 [14.8, 23.0]	25.0 [21.3, 28.7]
Delhi	19.8 [17.3, 22.2]	18.7 [14.2, 23.1]	23.5 [18.9, 28.2]	20.2 [17.3, 23.0]
Bangalore	27.4 [25.7, 29.1]	25.7 [23.1, 28.2]	27.6 [24.4, 30.9]	–

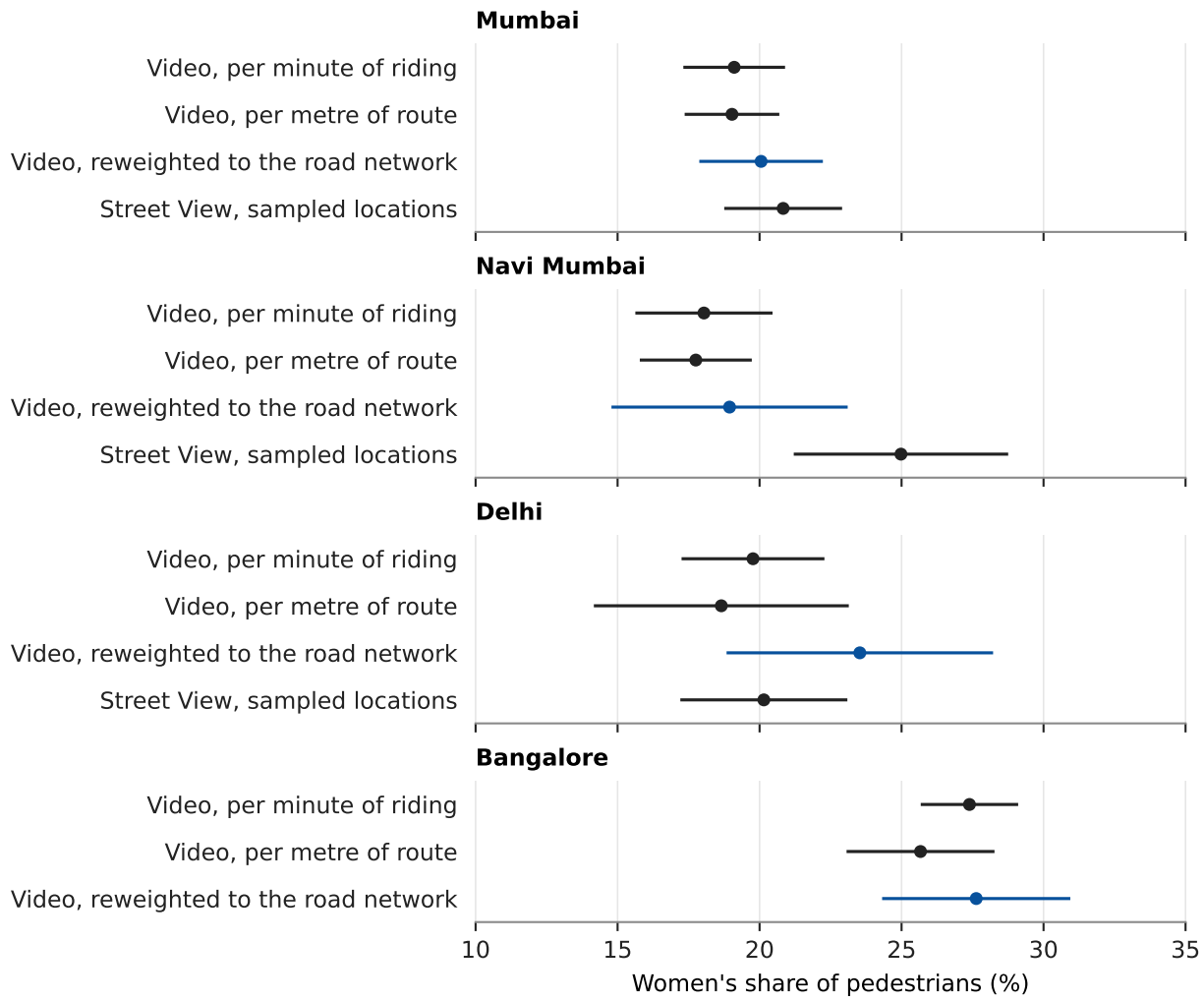


Figure 2: Women's share of pedestrians under each weighting of the video frames, and from Street View. Points are estimates and lines 95% confidence intervals, computed as in Table 3. The estimate reweighted to the road network is in blue.

Counted per minute of riding, women are 19.1% of pedestrians in Mumbai, 18.0% in Navi Mumbai, 19.8% in Delhi, and 27.4% in Bangalore. Weighting by speed lowers the Delhi and Bangalore estimates by one to two points, as the simulation predicts when faster roads have fewer women. Reweighting to the road network raises them again: to 23.5% in Delhi and 27.6% in Bangalore. In Delhi the reweighted estimate is 23.5% against 19.8% at face value. The adjustment matters most there because the Delhi route departs most from the sample (Figure 1) and because residential streets, where the share of women is highest, are the ones the route under-represents. In Mumbai, where the sample itself put more points on major roads, the three estimates differ by a point.

The adjustments also change how much the data can say. The reweighted intervals are

nearly twice as wide as the unadjusted ones in Delhi and Bangalore, because a few hundred residential frames now carry most of the weight. The clustered standard errors are between 0.6 and 1.8 times the ones that treat frames as independent.

Whatever the weighting, women are a fifth to a quarter of the pedestrians on these streets, and the video and Street View estimates agree to within their intervals in Mumbai and Delhi. They are furthest apart in Navi Mumbai, where Street View gives 25.0% and video at most 18.9%, though the reweighted video interval is wide enough to overlap Street View's. The two sources differ in year, hour, and the roads they cover, and we cannot say which difference is responsible. The location mean L is within a point of R in every city, so crowded places are not notably more male than quiet ones.

6.4 Noise in Delhi

Between 28 April and 16 May 2026 a rider carried a sound meter along 56 of the 85 Delhi itineraries, on 15 days. At stops along each itinerary the rider recorded the lowest and highest level the meter showed during the reading: 381 readings at planned stops and 81 more while halted in traffic. We do not know how the 56 itineraries ridden were chosen from the 85; they are spread across the city much as the rest are.

The readings are a route sample. Half of the planned-stop readings (50%) are on residential roads, against 85% of sampled points. Table 4 reports the readings as taken and reweighted to the sample's road classes, with itineraries as clusters.

Table 4: Street noise in Delhi from sound-meter readings, with 95% confidence intervals. Each reading gives the minimum and maximum level during the reading. "As read" weights planned-stop readings equally; "road network" post-stratifies them to the road classes of the drawn sample. Traffic stops are readings taken while halted in traffic and are not part of the other columns. Intervals treat itineraries as clusters. Every traffic-stop reading had a minimum above 55 dB, so that cell has no interval.

	Planned stops, as read	Planned stops, road network	Traffic stops
Mean minimum (dB)	64.5 [63.2, 65.9]	62.4 [61.2, 63.6]	71.5 [69.9, 73.1]
Mean maximum (dB)	82.3 [79.8, 84.8]	78.3 [76.3, 80.3]	94.8 [91.8, 97.8]
Minimum at or above 55 dB (%)	87.4 [82.9, 91.9]	83.5 [77.4, 89.6]	100.0
Maximum at or above 100 dB (%)	14.4 [9.5, 19.4]	8.1 [4.9, 11.2]	33.3 [23.3, 43.4]

Reweighted to the road network, the mean minimum is 62.4 dB and the mean maximum 78.3 dB. In 83.5% of readings even the quietest moment is at or above 55 dB, the daytime standard for residential areas. The reweighting matters most for the loudest moments: 14.4% of readings as taken peak at or above 100 dB, but 8.1% once residential streets get their due weight. Readings in traffic are louder, with a mean maximum of 94.8 dB and a third peaking

at or above 100 dB. They describe a condition, being stopped in traffic, and not a sample of streets, so we keep them apart.

A spot reading of a few seconds is not the day-long average that the standard is written for, and we did not calibrate the meter against a reference instrument. The comparison with 55 dB locates the readings; it is not a finding of legal exceedance.

6.5 Air quality in Delhi

On 19 of the itineraries the rider also read a hand-held air monitor, 98 times in all. Mean PM2.5 was 118 μg/m^3 (95% interval 89 to 146, itineraries as clusters). We report it as read. With 98 readings, the adjustments above would rest on a handful of readings per road class, and the interval is already wide. This application shows the limit of a small route sample more than it establishes a level.

7 Ethics

The imagery here shows people in public who did not agree to be recorded, and several features of the work are meant to limit what that costs them.

What is recorded about people. Coders count people and record apparent gender from clothing, hair, and build. No face recognition is used, no individual is identified or tracked across frames, and the analysis data contain counts only. Apparent gender is a coder's judgment about appearance. It imposes two categories on people who may not fit them.

What is kept and released. Video and full-resolution frames can identify people and are not released. Coders saw reduced-resolution frames. The public data are counts with a location and a time. Street View images are blurred by Google before we receive them, and we release no images from either source.

Location and the field team. The rider's routes are in the data as the positions of frames and readings. We checked that each day's first and last positions are scattered across the city and do not mark a home. The rider's route plan, which does contain a home address, is not released. Riding a motorbike in city traffic for weeks carries real risk to the rider, which a study of this kind should weigh against what it learns.

What the method could be used for. A procedure that samples streets and counts who is on them could be turned to watching particular places or groups. Probability sampling pulls the other way: it spreads observation thinly across a city and makes no place the target. We would not extend the method to identify individuals, to follow people, or to map the presence of a group in order to act against it.

Platform terms. Street View images are used under Google's terms, which restrict storing and redistributing images; we store counts and metadata.

Review. The video study was reviewed by the University of Southern California Institutional Review Board and deemed exempt (UP-25-00545).

8 Beyond Streets

Nothing above depends on the unit being a street. The frame can be any list of places that can be enumerated and reached: grid cells, parcels, transit stops, polling stations, water points. The two designs carry over. If the instrument observes the drawn place, the draw's probabilities apply. If the instrument observes the way to the place, the observations sample the way, and what must be asked is what governs how often each kind of place is seen (time, distance, access) and whether a recorded quantity, as speed was here, can undo it.

Roads are a convenient case because one object serves three purposes: the road network is the frame, the means of travel, and, for many outcomes, the thing measured. That convenience is also why route samples of roads go wrong in a patterned way. Travel concentrates on the roads built for travel.

9 Limitations

Road class is a coarse account of how routes differ from the network. Within residential streets, a route uses those that connect, and connecting streets may be busier than dead ends. The simulation does not test this, and our data cannot.

The road-class shares used for reweighting come from the drawn sample of 1,000 points and have their own sampling error, which the intervals ignore. In Delhi only 4% of sampled points are on secondary or larger roads, so that share is itself imprecise.

Fieldwork covered daytime on some days of some months. Every estimate is about those hours. Who is on the street at night, and how loud the street is then, are not measured.

The number of independent units is small for the video (9 to 14 days per city), and intervals with so few clusters are too narrow by an amount we have not estimated.

Several things were not recorded when the data were collected and limit what can be claimed: the size of the frame, how coded Street View locations were chosen, and how the itineraries ridden with the sound meter were chosen. The sampler now saves each segment's inclusion probability, stratum, and length with the sample.

10 Conclusion

A random draw from the road network gives street-level measurement a population to describe. Whether the measurements describe it depends on a second thing, which is easy to overlook: whether the instrument observed the units that were drawn. When it did, as with images requested at sampled segments, the usual survey estimators apply and the work is in reporting coverage. When the instrument rode between the units, the data describe the ride. In our Delhi fieldwork the ride spent 42% of its frames on large roads that make up 4% of the sample.

Two recorded quantities repair much of this. The rider's speed converts a camera's time-based sample into a distance-based one, and the drawn sample's road classes say how to reweight the route to the network. Neither is enough alone, and speed alone makes matters worse. Together they recover the truth in simulation, and in the data they move estimates by amounts that matter: several points in the share of women on Delhi's streets, and nearly half of the share of very loud noise readings.

For new studies the simpler course is to keep the two designs apart. Observe at the sampled points when the question is about the network. Record continuously when the question is about travel, and say that the population is the route. The same sample can also support comparisons over time or between places if the protocol is repeated, though we have not done that here.

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